SHINYA **YAMANAKA**

By "reprogramming" mature cells, Yamanaka was able to turn them back into stem cells.

In 2006, Yamanaka (1962–) introduced 24 specific genes into adult skin cells in mice and made them revert to an immature stem-cell state. He also showed that these cells were "pluripotent," meaning they can become any type of cell. Previously, it was thought that only embryonic stem cells were pluripotent. For his "induced pluripotent stem cells," or iPS cells, Yamanaka jointly received the Nobel Prize in Physiology or Medicine in 2012.

paper that referred to the "radiation sensitivity of normal mouse bone marrow cells." It was largely ignored.

In 1963, Till, McCulloch, and Becker published further results in the journal *Nature*. This time, they got attention. They went on to publish even more studies and clearly showed that bone marrow contains special cells that can reproduce and also develop into other types of cells, with specialized functions. We now know that, in adults, stem cells exist in many organs and tissues; this is how the body heals or regenerates damaged tissue. Their discovery was a vital advance in biological science and has huge potential for treating disease.

***Stem cells exist** in the body at all stages of a person's life. Embryonic stem cells can develop into any type of cell. Adult stem cells are more specialized—although recent research suggests they may be more flexible than once thought.*

AFTER THE **MARROW TRANSPLANTS,** THE **MICE** WERE **EXAMINED** FOR **10** TO **11**

DAYS

EACH **MOUSE SPLEEN NODULE** CORRESPONDED TO **10,000** BONE MARROW CELLS

STEM CELL RESEARCH **ACCELERATED** THE DEVELOPMENT OF **BONE MARROW TRANSPLANTS**